

Detailed Data Preprocessing Report

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Project Background

The AI-Based Personalized Diet and Recovery Planner project uses synthetic healthcare datasets to simulate patient recovery, dietary monitoring, and health analytics workflows. Raw datasets required extensive preprocessing before machine learning implementation.

Preprocessing Objectives

The preprocessing pipeline aimed to improve data quality, remove inconsistencies, validate relationships between datasets, standardize medical values, and prepare healthcare records for predictive analytics.

Dataset Inspection

Both datasets were initially inspected to validate column structures, identify relationships, understand feature distributions, and evaluate healthcare-specific attributes such as blood pressure, blood sugar, medication adherence, and symptom tracking.

Duplicate Record Validation

Duplicate records were analyzed using `duplicated()` functions and relational checks on `patient_id` and `record_id` fields. Duplicate rows were removed to preserve relational integrity and prevent biased analytical outcomes.

Data Type Validation

Date columns were converted into datetime formats. Numerical columns such as weight, calories, protein intake, blood sugar levels, and sleep hours were validated for numerical consistency.

Categorical Standardization

Categorical values were standardized by removing whitespace inconsistencies, correcting capitalization issues, and normalizing category names across datasets. This improved grouping accuracy during analytics.

Blood Pressure Feature Extraction

The blood_pressure field was split into systolic_bp and diastolic_bp numerical columns to improve statistical analysis and machine learning compatibility.

Range Validation

Healthcare-specific validation rules were applied to ensure realistic values. Age, heart rate, blood sugar, blood pressure, sleep duration, and calorie intake were validated against medically acceptable ranges.

Synthetic Data Expansion

Additional synthetic patient and tracking records were generated using disease-aware logic, progressive recovery simulation, and doctor-style clinical notes to improve dataset diversity.

Pipeline Summary

The final preprocessing pipeline integrated dataset inspection, missing value handling, duplicate validation, datatype correction, categorical normalization, feature extraction, and synthetic healthcare data generation.